AI SYNERGY HACKATHON 2026 · ROUND 1 · IMPLEMENTATION PLAN

LoopBack — Basic SRS

Software requirements and version-1 build plan for the OPD follow-up register: structured capture, missed-visit detection, and a prioritised, human-run recovery worklist.

Case 03 · Healthcare **Team** KitKat **Scope** Version 1 (Build Round slice)

01 Users and roles

Role	Uses	Can do
OPD nurse	Follow-up desk, nurse review	Confirm follow-ups; record contact & consent; see the clinician's note; close any case with a reason.
Records staff	Worklist, registration	Send reminders/texts; call; log outcomes; escalate; close routine & important cases with a reason.
ASHA worker	ASHA visit list	Log home-visit outcomes; escalate.
Hospital admin	Today page, audit trail	Read only.
Clinician	OPD slip (writes instruction, ticks importance)	Settle anything flagged "decide".

02 Inputs and outputs

Inputs

- Clinician's instruction (free text), department, visit date, importance tick
- Patient registration details
- Contact details & per-route consent, from the desk
- Registration check-ins
- Call and home-visit outcomes

Outputs

- A confirmed follow-up record
- A bilingual patient slip (हिंदी + English)
- Reminder & missed-visit texts (fixed templates)
- An ordered recovery worklist per lane
- Today-page figures, fairness check, and audit trail

03 Workflow

One record moves through four steps; a person confirms every decision.

Step	System proposes	Person decides
1 Capture	Reads the instruction → proposed due date + bring-list; flags the unreadable.	Nurse confirms or corrects every field; clinician settles flags.

Step	System proposes	Person decides
2 Check contact	Pre-fills contact form; fixed household-safe wording.	Patient gives consent per route; nurse records; slip printed.
3 Track	Date arithmetic: due / missed / returned; reminder 2 days before.	Registration checks patients in.
4 Recover	Ranks worklist (importance → days overdue); reason on every row.	Staff work top-down, call on the agreed route, log outcome & miss reason.

04 Functional requirements

ID	Requirement
FR-1	The desk lists patients leaving OPD with the instruction and a proposed reading. **Nothing is saved until the nurse confirms.**
FR-2	The reader proposes a due date from intervals (shorthand, Hinglish, Hindi), explicit dates and weekdays; ignores medicine durations; moves the date to the department's next OPD day.
FR-3	The reader **never invents a date**. SOS, as-needed, event-based, conditional and contradictory instructions are flagged for a person to decide.
FR-4	Importance (routine / important / must-not-miss) is set by the clinician and recorded by the nurse. The system only points out emphasis words.
FR-5	A follow-up is *missed* when its window closes with no registration visit. A *return* is recorded only from registration.
FR-6	Reminder texts go two days before the due date, only to patients who agreed to texts.
FR-7	Misses are listed in one order by a published points rule. Phone access and distance change the **route**, never the **rank**.
FR-8	Contact uses only routes the patient agreed to; otherwise the patient is flagged for registration.
FR-9	Texts use fixed templates that never name the department, tests, diagnosis or the clinician's words. Shared phones get household wording.
FR-10	Staff log every attempt with an outcome. Clinical questions go to a nurse. Retries are scheduled.
FR-11	No follow-up is closed without a recorded reason. Must-not-miss closures and declined/deceased/clinician-cancelled need a nurse. Only a registration visit resolves a case automatically.
FR-12	The optional Gemini reader receives only scrubbed instruction text. Its answer is kept only where it quotes the instruction; disagreements go to the nurse.
FR-13	Every action, and every view of a full phone number or the clinician's note, is written to the audit trail with date and role.
FR-14	The Today page shows open and uncontacted misses, on-time and recovery rates, the fairness check by contact group, and how often nurses corrected the reader.

05 AI and human responsibilities

Step	System (AI / rules)	People
Reading the instruction	Rule reader proposes; Gemini optionally cross-checks; both flag what they cannot read.	Nurse confirms or corrects every field; clinician settles flags.
Importance	Points out emphasis words only.	Clinician sets; nurse records.
Missed / returned	Date arithmetic; registration feed.	Registration checks patients in.
Order of work	Points rule, reason on every row.	Staff work from the top; can escalate.
Contact	First consented route; fixed message wording.	Patient consents; staff call and text.
Clinical questions	**Never answered.**	Nurse or clinician.
Closing a case	Never, except a registration visit.	Records or nurse, always with a reason.

06 Constraints

- No diagnosis, triage, treatment change or medical advice anywhere in the software.
- Works **offline**. The rule reader runs in the browser; Gemini is optional and off by default.
- Consent per contact route, recorded with date and role; nothing pre-ticked.
- **No patient identifiers** sent to any AI service.
- Due dates respect each department's OPD days and fixed public holidays.

07 Failure handling — missing, conflicting or unreliable information

Situation	Behaviour
No time in the instruction	No date proposed; flagged **`DECIDE`**; nurse asks the clinician or records "no fixed date" with a reason.
Two different times ("1 wk / 10 days")	Both shown; nothing pre-selected.
"If fever persists", "after completing ATT", "2 weeks after surgery"	Flagged; a person decides whether and when to track.
"6/12" could be a date or six months	Read as six months, flagged to check.
Medicine durations ("x 14 days")	Set aside, shown as such — never taken as the follow-up time.
Date on a day the clinic does not run	Moved to the next OPD day, with the reason shown.
Shared phone	Household wording only; privacy script for calls.
Wrong number	Marked unreachable; next consented route used.
No consented route	Registration alert; must-not-miss goes to a nurse.
Gemini unavailable, slow, over quota or ungrounded	Rule reading stands; reason shown; ungrounded parts discarded.

08 Acceptance criteria

16 checks run in `npm test` and on the in-app *About* page — each can fail.

ID	Check	Covers
AT-01	Shorthand becomes a proposed date; nothing saves until the nurse confirms.	FR-1,2
AT-02	No date is invented for SOS.	FR-3
AT-03	Contradictory instructions are shown side by side, not resolved.	FR-3
AT-04	Medicine durations are not taken as the follow-up time.	FR-2
AT-05	Conditional and event-based returns need a human decision.	FR-3
AT-06	Importance is never set by the system.	FR-4
AT-07	No message names the department, a test or the clinician's words.	FR-9
AT-08	Reachability changes the route, never the rank.	FR-7
AT-09	Must-not-miss comes first; routine patients rise as they wait.	FR-7
AT-10	Nothing is closed silently.	FR-11
AT-11	The clinician's note is visible to nurses only.	Privacy
AT-12	"Missed" is date arithmetic; "returned" comes from registration.	FR-5
AT-13	Due dates land on days the OPD runs.	FR-2
AT-14	Gemini's answer is kept only where it quotes the instruction.	FR-12
AT-15	Identifiers are removed before any text goes to Gemini.	FR-12
AT-16	Every contact route needs recorded consent.	FR-8

09 Version 1 exclusions

V1 deliberately does **not**:

- Diagnose, triage, change treatment or give medical advice.
- Predict who will miss, or judge clinical urgency.
- Contact anyone without recorded consent for that route — texts go through a **simulated** gateway.
- Close a follow-up by itself, other than a return seen at registration.
- Send names, phone numbers or registration numbers to any AI service.
- Connect to the hospital information system, or book transport.

10 How we will know it works

Measure	Definition	Baseline / target
Main outcome	Follow-ups returned within the window.	Case baseline 72%; report the change, no promised number.
Recovery	Misses that return within 30 days of the window closing.	Measured from the first pilot week.
Process	Confirmed date & checked contact at the desk; days from miss to first contact.	Never-contacted must-not-miss misses > 2 working days: zero.
Harm / exclusion	On-time & recovery gaps between own-phone, shared-phone and unreachable patients (and by distance).	Review the design if any gap exceeds 10 points.
Privacy	Texts revealing clinic, test or diagnosis.	Zero (AT-07).
Reader quality	Nurse corrections per field; Gemini disagreement rate.	Tracked weekly.

Comparison: four weeks before against four weeks after, in one OPD (Medicine), from registration records.

11 Risks and mitigations

Risk	Mitigation
Nurses accept proposals without reading them	Decide-flags block auto-fill; corrections tracked; audit trail.
The reader misses a phrasing used locally	Flags instead of guesses; Gemini as optional second reader; vocabulary extended from correction data.
Shared phones reveal sensitive visits	Household wording, no clinic/test names, privacy script, AT-07.
Hard-to-reach patients fall further behind	Route by consent, not rank; ASHA lane; fairness check on the Today page.
Free-tier AI used with real patient text	Off by default; identifiers scrubbed; simulated-data only; production needs a paid tier or on-premise model under a data agreement.

12 Build Round plan

1. Import a de-identified sample of real follow-up instructions; measure reader accuracy and extend the vocabulary.
2. Replace the simulated registration feed with a CSV or HL7 import from the hospital’s registration system.
3. Connect a real SMS gateway behind the same templates and consent checks.
4. Add a printed or WhatsApp-friendly ASHA visit list.
5. Run the four-week before/after comparison in one OPD.

> **Declarations.** Every patient, phone number, place and outcome is simulated from a fixed seed; proportions follow the case file and are not real statistics. Reused components: React, Vite, TypeScript, Vitest, Testing Library, Fontsource IBM Plex fonts (SIL OFL), and the optional `@google/genai` client. No templates or UI kits.

References — [1] Gurol-Urganci I, et al. *Cochrane Database Syst Rev* 2013;(12):CD007458. [2] Digital Personal Data Protection Act 2023, s.6. [3] Gemini API Additional Terms of Service, ai.google.dev/gemini-api/terms.